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LINEAR ACTUATOR

Abstract

A linear actuator includes a housing; a motor module that includes a motor; and a drivetrain module coupled to the motor module. The drivetrain module includes a ball screw coupled to the motor, a ball nut coupled to and configured to linearly ride on the ball screw during rotation of the ball screw, a rod coupled to the ball nut and configured to extend or retract from the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, and an anti-rotation bushing coupled to the rod and including at least one slot configured to engage with and linearly ride on a key coupled to an inner surface of the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw. The anti-rotation bushing is further configured to restrain the rod against rotation during extension and retraction.

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Background/Summary

TECHNICAL FIELD

[0002] The present disclosure describes example implementations of a linear actuator and, more particularly, a linear ball screw actuator used in a humanoid robot.

BACKGROUND

[0003] Linear actuators are used in many different technologies in order to move a load or object bi-directionally in a linear fashion. Many linear actuators have a relatively large weight to performance ratio, i.e., they are heavy given their output linear force. Further, many linear actuators use a leadscrew as a drivetrain, which often results in high friction. For these reasons and others, conventional linear actuators are often unsuitable for mobile robots, such as quadruped robots and humanoid robots.

SUMMARY

[0004] In an example implementation, a linear actuator includes a housing; a motor module that includes a motor; and a drivetrain module coupled to the motor module. The drivetrain module includes a ball screw coupled to the motor, a ball nut coupled to and configured to linearly ride on the ball screw during rotation of the ball screw, a rod coupled to the ball nut and configured to extend or retract from the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, and an anti-rotation bushing coupled to the rod and including at least one slot configured to engage with and linearly ride on a key coupled to an inner surface of the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw. The anti-rotation bushing is further configured to restrain the rod against rotation during extension and retraction.

[0005] In an aspect combinable with the example implementation, the motor includes a direct current (DC) motor.

[0006] In another aspect combinable with the any of the previous aspects, the motor includes a direct current (DC) brushless and frameless motor.

[0007] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes a bore configured to receive the rod.

[0008] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes at least one groove formed on an inner radial surface of the anti-rotation bushing and configured to engage the rod.

[0009] Another aspect combinable with the any of the previous aspects further includes a sealed angular contact bearing positioned in the housing between the motor module and the drivetrain module.

[0010] Another aspect combinable with the any of the previous aspects further includes a first bumper positioned in the housing between the motor module and the drivetrain module.

[0011] Another aspect combinable with the any of the previous aspects further includes a second bumper positioned in the housing at an end of the drivetrain module opposite the motor module.

[0012] In another aspect combinable with the any of the previous aspects, each of the first and

second bumpers include urethane bumpers.

[0013] In another aspect combinable with the any of the previous aspects, the first bumper includes a first stop that defines a retracted position of the rod when the ball nut is adjacent or abutting the first bumper.

[0014] In another aspect combinable with the any of the previous aspects, the second bumper includes a second stop that defines an extended position of the rod when the ball nut is adjacent or abutting the second bumper.

[0015] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes a rigid, non-metallic material.

[0016] In another aspect combinable with the any of the previous aspects, the rigid, non-metallic material includes a plastic material.

[0017] Another aspect combinable with the any of the previous aspects further includes an encoder configured to determine a number of rotations of the motor.

[0018] In another aspect combinable with the any of the previous aspects, a gear ratio is between 40:1 and 50:1.

[0019] In another aspect combinable with the any of the previous aspects, a gear ratio is no more than 100:1.

[0020] In another example implementation, a humanoid robot includes at least one body assembly that includes at least one body joint assembly; and at least one pair of linear actuators that form the at least one body joint, the at least one pair of linear actuators configured to operate in combination to adjust the at least one body joint assembly in two degrees of freedom through differential linear actuation. Each of the linear actuators includes a motor module that includes a motor; and a drivetrain module coupled to the motor module. The drivetrain module includes a ball screw coupled to the motor, a ball nut coupled to and configured to linearly ride on the ball screw during rotation of the ball screw, a rod coupled to the ball nut and configured to extend or retract from the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, and an anti-rotation bushing coupled to the rod and including at least one slot configured to engage with and linearly ride on a key coupled to an inner surface of the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw. The anti-rotation bushing is further configured to restrain the rod against rotation during extension and retraction.

[0021] In an aspect combinable with the example implementation, the at least one body assembly is an upper body assembly, and the at least one body joint assembly includes at least one of a shoulder assembly, an upper arm assembly, or a lower arm assembly.

[0022] In another aspect combinable with the any of the previous aspects, the at least one body assembly is a lower body assembly, and the at least one body joint assembly includes at least one of a hip assembly, an upper leg assembly, or a lower leg assembly.

[0023] In another aspect combinable with the any of the previous aspects, the body joint assembly includes the upper leg assembly, and the at least one pair of joint linear actuators includes a first linear actuator of a first size; a second linear actuator of the first size; and a third linear actuator of a second size larger than the first size.

[0024] In another aspect combinable with the any of the previous aspects, the motor includes a direct current (DC) motor.

[0025] In another aspect combinable with the any of the previous aspects, the motor includes a direct current (DC) brushless and frameless motor.

[0026] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes a bore configured to receive the rod.

[0027] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes at least one groove formed on an inner radial surface of the anti-rotation bushing and configured to engage the rod.

[0028] In another aspect combinable with the any of the previous aspects, each of the linear

actuators further includes a sealed angular contact bearing positioned in the housing between the motor module and the drivetrain module.

[0029] In another aspect combinable with the any of the previous aspects, each of the linear actuators further includes a first bumper positioned in the housing between the motor module and the drivetrain module.

[0030] In another aspect combinable with the any of the previous aspects, each of the linear actuators further includes a second bumper positioned in the housing at an end of the drivetrain module opposite the motor module.

[0031] In another aspect combinable with the any of the previous aspects, each of the first and second bumpers include urethane bumpers.

[0032] In another aspect combinable with the any of the previous aspects, the first bumper includes a first stop that defines a retracted position of the rod when the ball nut is adjacent or abutting the first bumper.

[0033] In another aspect combinable with the any of the previous aspects, the second bumper includes a second stop that defines an extended position of the rod when the ball nut is adjacent or abutting the second bumper.

[0034] In another aspect combinable with the any of the previous aspects, the anti-rotation bushing includes a rigid, non-metallic material.

[0035] In another aspect combinable with the any of the previous aspects, the rigid, non-metallic material includes a plastic material.

[0036] In another aspect combinable with the any of the previous aspects, each of the linear actuators further includes an encoder configured to determine a number of rotations of the motor.

[0037] In another aspect combinable with the any of the previous aspects, a gear ratio of each of the linear actuators is between 40:1 and 50:1.

[0038] In another aspect combinable with the any of the previous aspects, a gear ratio of each of the linear actuators is no more than 100:1.

[0039] Implementations of a linear actuator according to the present disclosure can include one, some, or all of the following features. For example, linear actuators according to the present disclosure are optimized for low friction, low backlash, and minimal mass, using aluminum and plastic structures for more efficient use in a mobile robot, such as a humanoid robot. As another example, linear actuators according to the present disclosure operate as “two force members” and are internally restrained from rotation and avoid heavy rotational bearings. As a further example, linear actuators according to the present disclosure are scalable for several sizes, gear ratios, and locations in a humanoid robot.

[0040] The details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] FIG. 1 is an example implementation of a linear ball screw actuator according to the present disclosure.

[0042] FIG. 2A is an axial cross-section of the linear ball screw actuator of FIG. 1 according to the present disclosure.

[0043] FIG. 2B is a schematic illustration that shows an axial cross-section of the linear ball screw actuator of FIG. 1 according to the present disclosure.

[0044] FIG. 3 is an isometric view of a portion of the linear ball screw actuator of FIG. 1 according

to the present disclosure.

[0045] FIG. 4 is an example implementation of a humanoid robot that uses an example implementation of a linear ball screw actuator according to the present disclosure.

DETAILED DESCRIPTION

[0046] FIG. 1 is an example implementation of a linear ball screw actuator **100** according to the present disclosure. While the linear ball screw actuator **100** can be used for a variety of mechanical tasks that require linear movement of a load or object that is coupled to the linear ball screw actuator **100**, in some aspects, the linear ball screw actuator **100** shown in FIG. 1 can be applied to mobile robots, such as humanoid robots. The application of the linear ball screw actuator **100** shown in FIG. 1 to a humanoid robot, such as use in upper body appendages (e.g., upper arms, forearms) and lower body appendages (e.g., upper legs, lower legs) can provide advantages not found in conventional linear actuators.

[0047] For example, greater operational efficiency of humanoid or mobile robot can depend on characteristics such as friction, backlash, reflected inertia, mass, and cost. Conventional linear actuators often are heavy for their performance and therefore not meant for mobile robotics. Further, many conventional linear actuators use a leadscrew as a drivetrain, which results in high friction that can reduce battery life, force transparency, back-driveability/human safety; each of these characteristics can be undesirable for mobile robots. Further, many conventional linear actuators use bulky and heavy roller bearings to prevent off-axis forces from damaging internal components of the actuator. In some cases, conventional linear actuators include a secondary transmission (such as a timing belt/pulley) between a motor and drivetrain, resulting in larger volume, larger mass, and generally higher gear ratios resulting in excess reflected inertia. Finally, conventional linear actuators often do not include anti-rotation features, thereby requiring external anti-rotation features upon integration into a machine such as mobile robot. This exclusion often prevents use in a MIMO (multi in, multi out) joint, where two actuators drive two joints differentially and thus require the use of spherical bearings at their ends, which cannot resist this rotation tendency.

[0048] The example implementation of the linear ball screw actuator **100** shown in FIG. 1 includes operational and structural characteristics that overcome the disadvantages of conventional linear actuators for mobile robots, such as humanoid robots. For example, the illustrated architecture of the linear ball screw actuator **100** is optimized for minimal mass, in part by using optimal materials such as aluminum and plastic (or other rigid, non-metallic) structures. The linear ball screw actuator **100** is designed to be a “two force member” by eliminating all or most off-axis loading that would conventionally require heavy linear bearings. Example implementations of the linear ball screw actuator **100** use a ball screw as a drivetrain, which can be low friction, low backlash, and low cost relative to a leadscrew. Further, the linear ball screw actuator **100** can use a brushless, frameless motor that is directly integrated with the ball screw drivetrain to minimize excess parts and secondary transmissions.

[0049] The linear ball screw actuator **100** can be integrated into a joint of a humanoid (or other mobile) robot with a lower than conventional gear ratio to minimize reflected inertia while also minimizing actuator mass. In some aspects, for example, lower body assembly joints of a humanoid robot may use the linear ball screw actuator **100** (e.g., in pairs) that have between a 40:1 and 50:1 gear ratio. Upper body assembly joints of a humanoid robot may use the linear ball screw actuator **100** (e.g., in pairs) that have up to a 100:1 gear ratio. In comparison, conventional linear actuators may have gear ratios of up to 200:1. The linear ball screw actuator **100** can be scaled for several sizes, gear ratios, and locations in a humanoid robot as well.

[0050] In order to provide more efficient linear movement, implementations of the linear ball screw actuator **100** can use a built-in anti-rotation mechanism by constraining a ball nut with a low-friction bushing that rides on anti-rotation keys of the housing of the linear ball screw actuator **100**. The illustrated linear ball screw actuator **100** can also use two separable modules

(motor/electronics and drivetrain) to improve production and increase ease of assembly. The linear ball screw actuator **100** can also use angular bearings to seal drivetrain grease from entering the motor/electronics module in this dual module architecture. Further, the linear ball screw actuator **100** can use a piston seal to seal drivetrain grease in the actuator and protect an inner volume of the linear ball screw actuator **100** from ingress.

[0051] As shown in FIG. **1**, the example implementation of the linear ball screw actuator **100** is comprised of a motor module **104** and drivetrain module **102** coupled together and enclosed in housing **106**. A rod end **108** extends from the motor module **104** and can be coupled, e.g., to a first point of a joint of a humanoid robot. A piston end **110** extends from the drivetrain module **102** and can be coupled, e.g., to a second point of a joint of a humanoid robot.

[0052] FIG. **2A** is an axial cross-section of the linear ball screw actuator **100** of FIG. **1**. FIG. **2B** is a schematic illustration that shows an axial cross-section of the linear ball screw actuator of FIG. **1**. As shown in FIGS. **2A** and **2B**, the housing **106** encloses the components of the motor module **104** and the drivetrain module **102**. Starting with the motor module **104**, this portion of the linear ball screw actuator **100** includes a motor **114** and contact bearings **116** to receive a ball screw **112** of the drivetrain module **102**. In some aspects, the motor **114** comprises a direct current (DC) motor, such as a brushless and frameless DC motor. The contact bearings **116**, in this example, can be face-to-face sealed, angular contact bearings that support the ball screw **112**, e.g., as it is forcibly rotated by the motor **114** to actuate the linear ball screw actuator **100**.

[0053] The motor module **104** further includes, in this example, an encoder **124** coupled to the motor **114**. The encoder **124**, in some aspects, is an incremental encoder for motor commutation such that a position of the linear ball screw actuator **100** (e.g., a position between fully retracted and fully extended) can be determined by a number of motor rotations of the motor **114**. In some further aspects, the encoder **124** can be a multi-turn absolute position encoder, in which the position of the linear ball screw actuator **100** can be absolutely determined based on motor operation.

[0054] As shown in FIGS. **2A** and **2B**, the drivetrain module **102** is coupled to the motor module **104** between the contact bearings **116** and a bumper **118a**, through each of which the ball screw **112** of the drivetrain module **102** extends. In some aspects, the bumper **118a** is a urethane bumper that minimizes vibration and impact within the linear ball screw actuator **100** during operation. A corresponding bumper **118b** (e.g., urethane bumper) is also positioned at an opposite end of the housing **106** in the drivetrain module **102** to further minimize vibration and impact during operation of the linear ball screw actuator **100**.

[0055] As shown in this example, the drivetrain module **102** (i.e., the drivetrain of the linear ball screw actuator **100**) includes the ball screw **112** and a ball nut **132** threaded onto the ball screw **112** and drivable by the ball screw **112** (e.g., by rotation thereof) to operate the linear ball screw actuator **100** between the retracted position (in which a distance between the rod end **108** and the piston end **110** is minimized) and an extended position (in which a distance between the rod end **108** and the piston end **110** is maximized). The ball nut **132**, threaded onto the ball screw **112**, is also coupled to rod **150**, which retracts and extends during actuation of the linear ball screw actuator **100** by the translation of rotational motion of the ball screw **112** to linear motion of the ball nut **132**.

[0056] The drivetrain module **102** further includes an anti-rotation bushing **122** that is coupled to the rod **150** (e.g., the rod **150** extends through the anti-rotation bushing **122**) and rides on one or more splines (or keys) **120** as the ball nut **132** moves linearly on the ball screw **112**. In some aspects, the anti-rotation bushing **122** provides a connection between the rod **150** and the housing **106** (on which the key(s) **120** are formed on or coupled to an interior surface), thereby preventing the rod **150** from rotating about a linear axis **101** of the linear ball screw actuator **100** (shown in FIG. **2B**) during linear motion of the rod **150** along the axis **101**. In some aspects, the anti-rotation bushing **122** can be made of a reduced friction material, such as plastic or other rigid, non-metal material in order to reduce friction during actuation of the linear ball screw actuator **100** and

eliminate heavier roller bearings (thereby allowing use of the linear ball screw actuator **100** in a MIMO joint).

[0057] Turning briefly to FIG. 3, this figure shows an isometric view of a portion of the linear ball screw actuator **100** of FIGS. 1, 2A, and 2B that includes the anti-rotation bushing **122**. As shown in FIG. 3, the anti-rotation bushing **122** is generally ring-shaped and includes one or more slots **140** formed on an outer radial surface to receive (and ride on during linear movement) the keys **120**. A bore **142** is formed in the anti-rotation bushing **122** to receive the rod **150** there through. In some aspects, as shown, one or more grooves **144** can be formed on an inner radial surface to engage keyed portions of the rod **150** (not shown). Thus, the anti-rotation bushing **122** can couple the rod **150** to the housing **106** (through the slots **140** engaging with the keys **120**) so prevent rotation of the rod **150** about the axis **101** during linear motion.

[0058] Turning back to FIGS. 2A and 2B, a support bushing **130** and rod seal **128** are positioned about the rod **150** toward the piston end **110** of the linear ball screw actuator **100**. The support bushing **130** provides further support to the rod **150** as it moves linear during actuation of the linear ball screw actuator **100**. The piston seal **128** can seal an interior volume of the housing **106** against lubricant leakage into an ambient environment and also seal against debris or moisture entering the housing **106** from the ambient environment. During operation of the linear ball screw actuator **100**, air or other fluid can escape the housing **106** through one or more vents **126** that open (e.g., due to an excess pressure in the housing **106**) to allow such release of fluid.

[0059] In an example operation of the linear ball screw actuator **100**, the motor **114** is activated, e.g., based on a command from a controller or sensor that reaches a threshold value. For example, when installed in a humanoid robot, a controller of the robot can signal the motor **114** to activate in order to achieve a specified movement of the robot (e.g., walking, reaching, pivoting, squatting, etc.). In some aspects, the signal can indicate a particular rotation of the motor **114** in order to either shorten (retract) or lengthen (extend) the linear ball screw actuator **100**. As the motor **114** is activated and begins to turn, it drives rotation of the ball screw **112**. The rotation of the ball screw **112** drives the ball nut **132** to rotate as well and linearly move along the ball screw **112**. The ball nut **132**, which is coupled to the rod **150**, drives the rod **150** in a linear fashion to extend from or retract into the housing **106** (depending on motor rotation direction).

[0060] In some aspects, bumper **118a** provides a “soft stop” to the ball nut **132** when moving in a “retracting” motion (i.e., toward the motor **114**), while bumper **118b** provides a “soft stop” to the ball nut **132** when moving in an “extending” motion (i.e., away the motor **114**). When the ball nut **132** is adjacent or abuts the bumper **118a**, the linear ball screw actuator **100** can be in a fully retracted position. When the ball nut **132** is adjacent or abuts the bumper **118b**, the linear ball screw actuator **100** can be in a fully extended position.

[0061] During linear motion of the rod **150** (and ball nut **132**) between the fully retracted position and fully extended position, little or no rotation of the rod **150** about axis **101** occurs. For example, the rod **150** is coupled to the housing **106** through engagement with the anti-rotation bushing **122** that is engaged with and rides on the keys **120**, which prevents rotation of the rod **150**. This MIMO linear movement of the linear ball screw actuator **100** without rotation can provide for advantages when used in a joint of a humanoid robot and, more particularly, when a pair (or triad) of linear ball screw actuators **100** are used to provide joint movement in a humanoid robot.

[0062] FIG. 4 is an example implementation of a humanoid robot **400** that uses an example implementation of a linear ball screw actuator according to the present disclosure. Humanoid robot **400** includes a head **402**, an upper body assembly **404**, and a lower body assembly **406** according to the present disclosure. Generally, humanoid robot **400** comprises a general purpose robot product that performs useful work in the real world (without the use of emotions) such as tasks that involve dangerous, hazardous, or even normal day-to-day tasks incapable (or capable) of being performed by a human being. Example tasks can include handling dangerous or hazardous materials (e.g., munitions, radioactive material, chemical material), loading and unloading (e.g.,

items or objects that are immovable or otherwise by a single or multiple human beings), or tasks performed in hazardous or dangerous environments.

[0063] In the present disclosure, the term “humanoid robot” can refer to a robot that is generally human in shape, e.g., with a torso, a trunk, two torso appendages (i.e., arms/hands), two trunk appendages (i.e., legs/feet), and a head or skull appendage. However, the term “humanoid robot” can also refer to a robot that resembles just a portion of a human, such as only a torso with torso appendages, or only a trunk with trunk appendages. In addition, the present disclosure describes aspects of a humanoid robot (such as, for example, pairs of linear actuators that form a joint assembly or part of an appendage and operate in combination to adjust the joint assembly or appendage in two degrees of freedom through differential linear actuation) that can be applied in non-humanoid robots, such as quadruped robots or otherwise.

[0064] Humanoid robot **400** can be autonomously controlled (untethered to any external control system) or human-controlled (e.g., tethered or wirelessly) to perform tasks. For example, humanoid robot **400** can perform useful work with mobility and kinematic movement that at least partially mimics that of a human being, and in spaces occupied by humans or not. The humanoid robot **400**, in some aspects, is designed for practical portability and movement and for mass production.

[0065] The humanoid robot **400** can perform at various levels of autonomy. For example, example implementations of the humanoid robot **400** can be enabled for untethered locomotion testing, with some limited manipulation capabilities. In some aspects, example implementations of the humanoid robot **400** can be configured for full manipulation and locomotion.

[0066] The upper body assembly **404** includes, for example, a torso assembly **420**, shoulder assemblies **408**, upper arm assemblies **410**, lower arm assemblies **412**, and a neck assembly **418**. Generally, one, some, or all of the illustrated assemblies comprise pairs of linear ball screw actuators **100** that work by differential linear actuation to adjust or move the assembly in at least two degrees of freedom (as explained in more detail herein).

[0067] The lower body assembly **406** includes a hip assembly **422** (that couples to the torso assembly **420**), upper leg assemblies **414**, lower leg assemblies **416**, and foot assemblies **424** (that in some aspects are part of the lower leg assemblies **416**). Generally, one, some, or all of the illustrated assemblies comprise pairs (or triads) of linear ball screw actuators **100** that work by differential linear actuation to adjust or move the assembly in at least two degrees of freedom (as explained in more detail herein). In some aspects, each upper leg assembly **414** includes a pair of linear ball screw actuators **100** of one size in combination with a third linear ball screw actuator **100** of another size (as described more fully herein).

[0068] The pairs (or triads) of linear ball screw actuators **100** described herein can be actuated to control movement through differential linear actuation in at least two degrees of freedom. For example, the shoulder assemblies **408** can provide for flexion and extension of the arms of the humanoid robot **400** (e.g., lifting the arm to the front and rear). The shoulder assemblies **408** can provide for abduction and adduction (AA) of the arms of the humanoid robot **400**. The upper arm assemblies **410** can provide for internal/external (IE) rotation of the arms of the humanoid robot **400**. The combination of the upper and lower arm assemblies **410** and **412** (e.g., in combination with a radial actuator in some aspects) can provide for flexion-extension (FE) of the lower arms of the humanoid robot **400**.

[0069] In example aspects of the humanoid robot **400**, the use (e.g., predominant use) of linear ball screw actuators **100** (which have desirable performance per cost and manufacturability compared to other forms of actuators) can provide for lower gear ratios (as compared to traditionally used actuators in GPRs), which enable more dynamic behaviors and increases system robustness to environmental collisions/interactions. The kinematic use of linear ball screw actuators **100** as shown can provide for ordering and mechanisms that enable the humanoid robot **400** to move in a fashion that achieves desired functional attributes such as payloads/torques, speeds, efficiency, human-like form factor, and also sufficient range of motion. The illustrated pairs of linear ball

screw actuators **100** also possess variable moment for appendages (e.g., legs and/or arms) that mimic the way human muscles work in nature.

[0070] In some aspects, each of the illustrated linear ball screw actuators **100** can be sized for expected or desired movement and/or force output (e.g., for lifting or otherwise asserting a force to provide for movement of the humanoid robot **400**, itself, or of another object). In some aspects, use each of the illustrated linear ball screw actuators **100** can use a motor paired with a ball screw (as previously described) for primary (and efficient) speed reduction with a low (e.g., between 10:1 and 50:1) gear ratio. In some aspects each of the illustrated linear ball screw actuators **100** can be a quasi-direct drive (QDD) actuator. As described, each pair of the illustrated linear ball screw actuators **100** can operate through differential linear actuation, such that one linear ball screw actuator **100** of the pair can be controlled (e.g., elongated or shortened) independently of the other linear ball screw actuator **100** of the pair.

[0071] As described, each of the illustrated pairs of linear ball screw actuators **100** can provide (through differential linear actuation) of at least two degrees of freedom of movement of a particular portion of the humanoid robot **400**. For example, two degrees of shoulder freedom of roll and yaw can be provided through differential linear actuation of the pair of linear ball screw actuators **100** of shoulder assemblies **408**. Two degrees of torso freedom of roll and pitch can be provided through differential linear actuation of a pair of linear ball screw actuators **100** of torso assembly **420**. Two degrees of ankle freedom of roll and pitch can be provided through differential linear actuation of the pair of linear ball screw actuators **100** of lower leg assemblies **416**. Two degrees of hip freedom of roll and pitch can be provided through differential linear actuation of the pair of (smaller) linear ball screw actuators **100** and a (larger) thigh linear ball screw actuator **100** of upper leg assemblies **414**.

[0072] The two degrees of freedom of particular assemblies of the humanoid robot **400** can have particular ranges of motions. For example, Table 1 shows example ranges of motions (ROM) for particular degrees of freedom (the values of which are for examples only):

TABLE-US-00001 TABLE 1 Deg. of Absolute Absolute Working Working Freedom ROM (+) ROM (-) ROM (+) ROM (-) Torso pitch 20 -80 15 -75 Torso roll 15 -15 10 -10 Torso yaw 50 -60 45 -45 Shoulder AA 10 -95 5 -90 Shoulder IE 30 -30 25 -25 Shoulder FE 35 -125 30 -120 Elbow FE 10 -150 5 -145 Hip IE 35 -65 30 -60 Hip AA -45 -20 40 -15 Hip FE 35 -110 30 -105 Knee FE 150 0 145 5 Ankle FE 45 -90 42.5 -87.5 Ankle IE 20 -45 15 -40

[0073] Each of the particular degrees of freedom shown in Table 1 can have a maximum continuous torque in a range of between about 20 to about 150 Nm. Each of the particular degrees of freedom shown in Table 1 can have a maximum peak torque in a range of between about 75 to about 500 Nm. Each of the particular degrees of freedom shown in Table 1 can have a minimum no-load speed in a range of between about 3 to about 12 radians/sec. Each of the particular degrees of freedom shown in Table 1 can have a maximum gear ratio in a range of between about 20 to about 170. The linear ball screw actuators **100** that provide for particular upper body degrees of freedom shown in Table 1 can have a maximum actuator reflected inertia in a range of between about 0.05 to about 2 kg*m.sup.2. The linear ball screw actuators **100** that provide for particular upper body degrees of freedom shown in Table 1 can have a minimum actuator reflected inertia in a range of between about 0.02 to about 1.5 kg*m.sup.2.

[0074] Another not shown in FIG. 4, certain components, such as motor controllers for linear and radial actuators and other control components include and/or are connected by wiring or cabling. One issue that makes some robots appear less human and not conform to a human envelope is excessive cabling that is visible outside of a robot's outer shell. This excessive wiring may also present a snag hazard. Example implementations of the humanoid robot **400** minimize external cabling by maintaining cabling internally or minimizing external cabling. Example implementations of the humanoid robot **400** also help maintain cabling within the human envelope without putting undue stress on the cabling. More particularly, example implementations of the

humanoid robot **400** can define wire paths across joints to minimize stress on both sides of the joint, which allows for no or minimum strain on the board connectors to which cabling connects. [0075] Cabling and board joints may experience undue stress when the ratio of cable path length change to total cable length is too high. Minimizing the cable path length change through the range of motion of a joint to total cable length can ensure that cable does not stretch and put unnecessary stress on the cable, connectors, or boards. Furthermore, bending cables with too sharp a radius can induce local stresses in the cable, which can propagate to apply stresses on the connectors or boards. Example implementations of the humanoid robot **400** can implement features to minimize cable path length change and maximize bend radius.

[0076] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what may be claimed, but rather as descriptions of features specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0077] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

[0078] A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, example operations, methods, or processes described herein may include more steps or fewer steps than those described. Further, the steps in such example operations, methods, or processes may be performed in different successions than that described or illustrated in the figures. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A linear actuator, comprising: a housing; a motor module that comprises a motor; and a drivetrain module coupled to the motor module and comprising: a ball screw coupled to the motor, a ball nut coupled to and configured to linearly ride on the ball screw during rotation of the ball screw, a rod coupled to the ball nut and configured to extend or retract from the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, and an anti-rotation bushing coupled to the rod and comprising at least one slot configured to engage with and linearly ride on a key coupled to an inner surface of the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, the anti-rotation bushing further configured to restrain the rod against rotation during extension and retraction, the anti-rotation bushing comprising a bore configured to receive the rod.
2. The linear actuator of claim 1, wherein the motor comprises a direct current (DC) motor.
3. The linear actuator of claim 1, wherein the motor comprises a direct current (DC) brushless and frameless motor.
4. (canceled)

5. The linear actuator of claim 1, wherein the anti-rotation bushing comprises at least one groove formed on an inner radial surface of the anti-rotation bushing and configured to engage the rod.
6. The linear actuator of claim 1, further comprising a sealed angular contact bearing positioned in the housing between the motor module and the drivetrain module.
7. The linear actuator of claim 1, further comprising a first bumper positioned at a first end of the housing adjacent the motor module.
8. The linear actuator of claim 7, further comprising a second bumper positioned at a second end of the housing at an end of the drivetrain module opposite the first end.
9. The linear actuator of claim 8, wherein each of the first and second bumpers comprise urethane bumpers.
10. The linear actuator of claim 8, wherein the first bumper comprises a first stop that defines a retracted position of the rod when the ball nut is adjacent or abutting the first bumper.
11. The linear actuator of claim 8, wherein the second bumper comprises a second stop that defines an extended position of the rod when the ball nut is adjacent or abutting the second bumper.
12. The linear actuator of claim 1, wherein the anti-rotation bushing comprises a rigid, non-metallic material.
13. The linear actuator of claim 12, wherein the rigid, non-metallic material comprises a plastic material.
14. The linear actuator of claim 1, further comprising an encoder configured to determine a number of rotations of the motor.
15. The linear actuator of claim 1, wherein a gear ratio is between 40:1 and 50:1.
16. The linear actuator of claim 1, wherein a gear ratio is no more than 100:1.
17. A humanoid robot, comprising: at least one body assembly that comprises at least one body joint assembly; and at least one pair of linear actuators that form the at least one body joint, the at least one pair of linear actuators configured to operate in combination to adjust the at least one body joint assembly in two degrees of freedom through differential linear actuation, each of the linear actuators comprising: a motor module that comprises a motor; and a drivetrain module coupled to the motor module and comprising: a ball screw coupled to the motor, a ball nut coupled to and configured to linearly ride on the ball screw during rotation of the ball screw, a rod coupled to the ball nut and configured to extend or retract from the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, and an anti-rotation bushing coupled to the rod and comprising at least one slot configured to engage with and linearly ride on a key coupled to an inner surface of the housing based on the ball nut linearly riding on the ball screw during rotation of the ball screw, the anti-rotation bushing further configured to restrain the rod against rotation during extension and retraction.
18. The humanoid robot of claim 17, wherein the at least one body assembly is an upper body assembly, and the at least one body joint assembly comprises at least one of a shoulder assembly, an upper arm assembly, or a lower arm assembly.
19. The humanoid robot of claim 17, wherein the at least one body assembly is a lower body assembly, and the at least one body joint assembly comprises at least one of a hip assembly, an upper leg assembly, or a lower leg assembly.
20. The humanoid robot of claim 19, wherein the body joint assembly comprises the upper leg assembly, and the at least one pair of joint linear actuators comprises: a first linear actuator of a first size; a second linear actuator of the first size; and a third linear actuator of a second size larger than the first size.
21. The humanoid robot of claim 17, wherein the motor comprises a direct current (DC) motor.
22. The humanoid robot of claim 17, wherein the motor comprises a direct current (DC) brushless and frameless motor.
23. The humanoid robot of claim 17, wherein the anti-rotation bushing comprises a bore configured to receive the rod.

- 24.** The humanoid robot of claim 23, wherein the anti-rotation bushing comprises at least one groove formed on an inner radial surface of the anti-rotation bushing and configured to engage the rod.
- 25.** The humanoid robot of claim 17, wherein each of the linear actuators further comprises a sealed angular contact bearing positioned in the housing between the motor module and the drivetrain module.
- 26.** The humanoid robot of claim 17, wherein each of the linear actuators further comprises a first bumper positioned at a first end of the housing adjacent the motor module.
- 27.** The humanoid robot of claim 26, wherein each of the linear actuators further comprises a second bumper positioned at a second end of the housing at an end of the drivetrain module opposite the first end.
- 28.** The humanoid robot of claim 27, wherein each of the first and second bumpers comprise urethane bumpers.
- 29.** The humanoid robot of claim 27, wherein the first bumper comprises a first stop that defines a retracted position of the rod when the ball nut is adjacent or abutting the first bumper.
- 30.** The humanoid robot of claim 27, wherein the second bumper comprises a second stop that defines an extended position of the rod when the ball nut is adjacent or abutting the second bumper.
- 31.** The humanoid robot of claim 17, wherein the anti-rotation bushing comprises a rigid, non-metallic material.
- 32.** The humanoid robot of claim 31, wherein the rigid, non-metallic material comprises a plastic material.
- 33.** The humanoid robot of claim 17, wherein each of the linear actuators further comprises an encoder configured to determine a number of rotations of the motor.
- 34.** The humanoid robot of claim 17, wherein a gear ratio of each of the linear actuators is between 40:1 and 50:1.
- 35.** The humanoid robot of claim 17, wherein a gear ratio of each of the linear actuators is no more than 100:1.
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